

Sensitivity of Antarctic sea level contribution to regularisation of the initialising inverse problem

Jonathan R Barnsley¹, Tamsin L Edwards¹, Alex Bradley¹, Stephen L Cornford², and Matt Trevers²

¹King's College London, Department of Geography, London, UK

²University of Bristol, Department of Geography, Bristol, UK

Correspondence: Jonathan R Barnsley (jonathan.barnsley@kcl.ac.uk)

Abstract. The ice sheet initial state is known to be a significant source of uncertainty in future sea level contribution, but this uncertainty is rarely explored outside of large-scale model intercomparison projects. Century-scale projections of the Antarctic Ice Sheet commonly generate an initial state by solving an inverse problem for basal friction and ice stiffness such that modelled surface ice velocities closely match observations. However, inverse problems are ill-posed and can absorb errors originating elsewhere in the model, for example in the bed topography or unresolved physics. While this yields realistic short-term behaviour, it may undermine the accuracy of long-term projections.

Here, we quantify how choices made in the inverse problem influence Antarctica's multi-century sea level contribution. Using the BISICLES ice sheet model, we generate an ensemble of initial states by perturbing two regularisation parameters, which control the smoothness of inferred basal friction and ice stiffness. We then project them forward to 2300 under both low and high emission scenarios. We find that sensitivity to regularisation of the ice-stiffness field is comparable to other commonly perturbed parameters on a century timescale but declines on longer timescales, whereas sensitivity to friction regularisation is consistently small — much lower than reported for an equivalent study of the Amundsen Sea Embayment. Our results suggest that the importance of these initialisation choices for sea level projections may depend on the model, domain, resolution, or timescale considered.

1 Introduction

The Antarctic Ice Sheet is the largest potential contributor to sea level rise, but its response to ongoing and future climate change remains highly uncertain (Fox-Kemper et al., 2021). This uncertainty is even more pronounced on timescales spanning multiple centuries, with projections to 2300 under high emissions ranging from near-zero to multiple metres of sea level contribution (Golledge et al., 2015; Bulthuis et al., 2019; DeConto et al., 2021; Greve et al., 2023; Seroussi et al., 2024; Coulon et al., 2025). Quantifying uncertainty in the timing and magnitude of sea level rise is important for climate adaptation strategies such as sea defenses, and underestimating uncertainty poses a risk to the resilience of coastal communities (Hinkel et al., 2019; Kopp et al., 2019). Ice sheet modelling studies that seek to produce usable projections of future sea level rise should therefore make efforts to include the dominant sources of uncertainty.

Uncertainty in the output of ice sheet models can be decomposed into parametric uncertainty, uncertainty in the climate forcing, and model structural uncertainty, which itself encompasses the form of parameterisations, the stress-balance approximation, the initial state, and much more. Most commonly, studies will consider uncertainty in the climate forcing, either by selecting numerous emission scenarios, climate models, or else through a climate indexing method (e.g. DeConto et al., 2021; Greve et al., 2023). Increasingly, studies are sampling a wide range of uncertain model parameters and parameterisations (Bulthuis et al., 2019; Hill et al., 2021; Coulon et al., 2024, 2025; Rosier et al., 2025). Model structural uncertainty, whilst touched on by some of these studies (notably Coulon et al., 2025), is largely explored through international projects such as the Ice Sheet Model Intercomparison Project (ISMIP6, Nowicki et al., 2016), and is the largest source of uncertainty in those experiments (Seroussi et al., 2023). The InitMIP branch of ISMIP6 was designed to specifically isolate uncertainty in the ice sheet initial state, showing that projected sea level contribution is highly sensitive to the choice of initialisation (Seroussi et al., 2019). However, outside of these large intercomparison projects, uncertainty in the ice sheet initial state is less well-explored, since producing multiple realistic initial states is often time consuming and computationally expensive.

The initial state refers to a set of spatial fields defining the physical condition of the ice sheet, which can include ice thickness, bedrock elevation, ice temperature or enthalpy, basal properties, damage, and others. Ideally, these would be similar to the modern ice sheet whilst producing short-term behaviour that resembles observed trends. However, this is not straightforward to achieve, mainly due to inconsistencies between datasets and the model physics. Initialisation methods that approach this problem can be broadly divided into two philosophies: model spin-ups and data assimilation, although some approaches combine elements of both (Pollard and DeConto, 2012; Le Clec'H et al., 2019). Here, we focus only on initialisation through data assimilation, which accounted for 7 of the 25 submissions to initMIP. Based on early work by MacAyeal (1992), this typically solves an inverse problem for basal friction and ice rheology parameters such that modelled ice velocities closely match observations. Approaches of this kind have been implemented into numerous ice sheet models and applied to the Antarctic Ice Sheet or its basins (Morlighem et al., 2013; Gagliardini et al., 2013; Cornford et al., 2015; Arthern et al., 2015; Hoffman et al., 2018; Rosier et al., 2025). Inverse methods can produce initial states with very good fits to observed dynamics. However, they are also ill-posed in that the data does not uniquely constrain the solution. Other equally plausible solutions to the inverse problem could yield different future sea level outcomes, but these are rarely explored. Furthermore, inverse problems can introduce compensating errors for deficiencies elsewhere in the model, which then influence future mass loss when the model is projected forward (Berends et al., 2023).

Standard approaches to the inverse problem minimise a loss function comprised of a model-observation misfit and regularisation term. This yields a single best estimate but gives no information about its associated uncertainty. For this, one would need to know the local curvature of the loss function about its minimum, which is described by the Hessian matrix. For practical ice sheet modelling purposes, the Hessian is too large to compute directly, but can be approximated via its low-rank structure (Petra et al., 2014). The resulting problem is still expensive to solve, but has allowed for the local approximation of the posterior distribution in idealised ice sheet problems (Petra et al., 2014; Babaniyi et al., 2021) or small realistic domains (Gopalan et al., 2021; Brinkerhoff, 2022). Isaac et al. (2015) perform a Bayesian inversion for basal friction over the Antarctic ice sheet by further assuming that the posterior is locally Gaussian about the optimal solution (equivalently: the loss function

is locally quadratic). However, this may not translate well to inverse problems that solve for both friction and rheology, since non-uniqueness of the solution could produce posterior geometry that is poorly represented by a Gaussian approximation. A Bayesian inversion for both friction and rheology has yet to be completed at a continent scale, motivating alternative methods that do not sample from the posterior, but nonetheless explore uncertainty in the solution to the inverse problem. The simplest approach is to produce an ensemble of projections in which the solved fields from a single inversion are variably scaled by a uniform multiplier (e.g. Nias et al., 2016; Bevan et al., 2023). These studies then downweight ensemble members with poor fits to observed dynamics using a post-hoc calibration. However, this only allows for uniform scaling and leaves the spatial pattern unchanged. Some studies perturb the bedrock elevation and solve inverse problems for each realisation (Wernecke et al., 2022; Krishna et al., 2026), though care must be taken to account for spatially autocorrelated errors in the topography dataset. A third approach is to treat hyperparameters within the inversion as uncertain parameters and perturb these in a traditional uncertainty quantification framework (Rosier et al., 2025). This allows for spatially-varying perturbations that are physically plausible by design.

Methods to solve the inverse problem vary between models (see e.g. Barnes et al., 2021), but most rely on some form of regularisation. If unregularised, the optimisation introduces short-wavelength, high-amplitude variability into the inferred fields, which overfit to noisy errors in the observed velocity. Regularisation imposes structure on the solution by penalising roughness and encouraging smooth solutions. However, the strength of regularisation is left largely up to the discretion of the user. An L-curve heuristic is commonly used to justify a choice of regularisation strength by finding the optimal compromise between solution fit and smoothness, characterised by the corner of the ‘L’ (Hansen, 1992). This does not guarantee, however, that the corner of the L-curve corresponds to anything physically meaningful, nor that the resulting fields are representative of real basal and rheological properties. The location of the L-curve corner has also been shown to depend on choices such as mesh resolution and parameterisations, which complicates its use as an objective criterion (Wolovick et al., 2023). Furthermore, Rosier et al. (2025) demonstrate that projected sea level contribution from the Amundsen Sea Sector can be highly sensitive to regularisation in the inverse problem as formalised by the Úa ice sheet model. This suggests that regularisation hyperparameters may be a route to exploring uncertainty in sea level projections arising from the inverse problem — and, by association, the initial state — but it is not yet clear how this result translates to other models or domains.

To further study uncertainties associated with the inverse problem, we use the BISICLES ice sheet model to assess the sensitivity of Antarctic sea level contribution to regularisation parameters in the inverse problem. We produce an ensemble of initial states by perturbing these parameters, alongside other commonly perturbed ice sheet model parameters, and project them forward to 2300 under a high-emission scenario. We find that regularisation parameters can be a moderate source of uncertainty in century-scale sea level projections, but that this sensitivity is likely not consistent across models, domains, and timescales. We discuss the implications of these results for future research efforts to explore initial state uncertainty in sea level projections.

2 Methods

2.1 BISICLES ice sheet model

BISICLES (Cornford et al., 2013) is a finite volume, vertically integrated ice sheet model, which has been used extensively for century-scale modelling of the Antarctic Ice Sheet (Cornford et al., 2015; Nias et al., 2019; Sun et al., 2020; Wernecke et al., 2022; Siahhaan et al., 2022; O’Neill et al., 2025). It uses the LIL2 approximation to Stokes flow (Schoof and Hindmarsh, 2010), which incorporates membrane stresses with a simplified treatment of vertical shear, allowing it to simulate a wide range of flow regimes in Antarctica with computational efficiency. It also features an adaptive mesh, which refines mesh resolution close to the grounding line and in regions of fast flow. Here, we run BISICLES with a base resolution of 8 km and 3 levels of refinement, up to a finest resolution of 1 km. This significantly improves the representation of grounding line dynamics compared to coarser grids (Cornford et al., 2016).

BISICLES solves the 2-dimensional LIL2 mass transport and stress balance equations outlined in Cornford et al. (2015). They feature a vertically integrated effective viscosity, $\phi h \bar{\mu}$, defined by:

$$\phi h \bar{\mu}(x, y) = \phi \int_{s-h}^s \mu(x, y, z) dz, \quad (1)$$

where h is ice thickness, s is surface elevation, and ϕ is an uncertain, spatially-varying coefficient that is determined by the inverse problem. Values of $\phi < 1$ correspond to softening of the ice and values $\phi > 1$ to strengthening, with a default value $\phi = 1$. The vertically varying effective viscosity, $\mu(x, y, z)$, is calculated using Glen’s flow law (Glen, 1955):

$$\mu(x, y, z) = \frac{1}{2A_0 A(T) \tau^{n-1}}, \quad (2)$$

where $A(T)$ is a temperature-dependent Paterson rate factor (Cuffey and Paterson, 2010), τ is the effective deviatoric stress, and the flow law exponent n is an uncertain parameter that we perturb in our experiment. The 3-dimensional temperature field is derived from a 100,000-year thermal spin-up described in Trevers et al. (2024). Previous studies have perturbed n after model initialisation by scaling $A(x, y)$ to ensure that the effective viscosity is unchanged everywhere (Getraer and Morlighem, 2025; Rosier et al., 2025). However, we follow (Martin et al., 2026) by fixing n before initialisation and introducing a spatially-uniform correction factor, $A_0 = 100,000^{(3-n)}$ to ensure that the effective viscosity is independent of n at a reference stress of 100 kPa. Either side of this value, the flow law diverges from the $n = 3$ case, but the effective viscosity is scaled by ϕ during the inverse problem so that the initial state matches observed ice velocities.

Basal sliding is governed by a regularised-Coulomb sliding law, which smoothly transitions between Weertman and Coulomb behaviour near to a reference velocity $u_0 = 300 \text{ m a}^{-1}$ (Joughin et al., 2019). As such, the slow-moving ice sheet interior stays in a power-law regime, whilst fast-flowing ice streams approach the perfectly plastic sliding limit. The basal shear stress, $\tau_b = -\alpha \mathbf{u}_b$, is given by:

$$\alpha = C |\mathbf{u}_b|^{m-1} \left(\frac{|\mathbf{u}_b|}{u_0} + 1 \right)^{-m}, \quad (3)$$

where the friction exponent m is an uncertain parameter that we perturb in our experiment and C is a spatially-varying friction coefficient that is also determined by the inverse problem. The form in (3) is equivalent to Joughin et al. (2019) with $m \rightarrow 1/m$ and $C \rightarrow Cu_0^m$, giving C the same units as in the standard power law. This performs similarly to other commonly used sliding laws for key glaciers in Antarctica (Barnes and Gudmundsson, 2022).

125 2.2 Model initialisation

We initialise BISICLES using an inverse method to solve for the basal friction coefficient, C , and the viscosity coefficient, ϕ , such that modelled surface velocities closely match observations. BISICLES uses conjugate gradient descent (Shewchuk, 1994) to minimise a cost function, J , defined by:

$$J = J_m + J_p, \quad (4)$$

130 where J_m is a function of the velocity misfit

$$J_m = \frac{1}{2} \int_{\Omega_V} (|\mathbf{u}| - |\mathbf{u}_{obs}|)^2 d\Omega_V, \quad (5)$$

and J_p is a Tikhonov penalty function

$$J_p = \frac{\alpha_C}{2} \int_{\Omega_V} |\nabla C|^2 d\Omega_V + \frac{\alpha_\phi}{2} \int_{\Omega_V} |\nabla \phi|^2 d\Omega_V, \quad (6)$$

135 where α_C and α_ϕ are regularisation parameters that respectively control the smoothness of C and ϕ , Ω_V is the domain over which velocities are observed, and $\mathbf{u}_{obs}$ are the surface velocities from Rignot et al. (2017). In our experiment, we treat both α_C and α_ϕ as uncertain parameters within a range of values (Section 2.4).

We solve the inverse problem using a linear sliding law ($\tau_b = -C_1 \mathbf{u}_b$) to simplify the stress balance, although this does come with a trade-off of more rounded, less well-defined L-curve corners (Wolovick et al., 2023). We then adjust C post-inversion such that basal drag (and therefore ice velocity) remains unchanged. This is done by scaling C over grounded ice such that

$$140 \quad C_m = C_1 |\mathbf{u}|^{1-m} \left(\frac{|\mathbf{u}|}{u_0} + 1 \right)^m. \quad (7)$$

Here, a minimum of 1 m a^{-1} is applied over $|\mathbf{u}(x, y)|$ to avoid unrealistically high friction in regions of very slow flow. This has the drawback that the initial velocity of ice in regions with $|\mathbf{u}| \ll 1$ may no longer be an optimal fit to observations, but only occurs in regions in which the observed velocity already poorly constrains basal friction (Isaac et al., 2015). The initial ice geometry is taken from BedMachine Antarctica v3 (Morlighem, 2022), the initial guess for ϕ is uniformly 1, and the initial
145 guess for C is calculated by equating basal drag to the driving stress. This yields

$$C_{init} = \rho_i g h |\nabla s| |\mathbf{u}|^{-1}, \quad (8)$$

where $\rho_i = 917 \text{ kg m}^{-3}$ is the density of ice and $g = 9.81 \text{ m s}^{-2}$ is gravitational acceleration. As is common with initialisation by inverse problem, we observe a short-term shock immediately after initialisation as the model adjusts to inconsistencies

between the geometry and model physics. We therefore allow a 15-year relaxation period, during which we periodically re-
 150 solve the inverse problem each year following Bevan et al. (2023). Throughout the relaxation, the thickness of floating ice
 shelves is held constant and we apply a constant background surface mass balance, derived by taking the 1980–2021 mean
 surface mass balance from a simulation by the MAR regional climate model, forced by ERA5-Interim data (Agosta et al.,
 2019) **NOTE: I got this dataset from Matt — Matt, do you know if this is the correct citation and have I described this
 correctly? Because Agosta et al. only seems to go up to 2015.** In total, we generate 7 initial states for our experiment by
 155 perturbing parameters used within the initialisation (see Section 2.4). These are then projected forward under a climate forcing
 described in Section 2.3.

2.3 Climate forcing

Following initialisation, we apply a time-varying surface and basal mass balance between an initial date in 2007 until 2300. The
 surface mass balance is composed of the background rate used in the relaxation, upon which we apply anomalies derived from
 160 the CESM2-WACCM climate model (Danabasoglu et al., 2020) under the SSP5-8.5 emissions scenario (Meinshausen et al.,
 2020). The anomalies are annual means with respect to the 1995–2014 climatology, calculated by subtracting evaporation,
 sublimation and runoff from precipitation, as in (Nowicki et al., 2020). CESM2-WACCM was chosen for its strong climate
 forcing to maximise the signal-to-noise ratio in our results, but its high climate sensitivity and the aggressive increase in
 emissions in SSP5-8.5 mean it is not necessarily representative of the most likely future climate (Hausfather, 2025). **TODO:**
 165 **ref for CESM2 climate sensitivity**

To compute a basal mass balance, we follow the method described by Jourdain et al. (2020), using the non-local quadratic
 parameterisation of ice shelf basal melt with respect to ocean thermal forcing.

$$m(x, y) = \gamma_0 \times \left(\frac{\rho_{\text{sw}} c_{\text{pw}}}{\rho_i L_f} \right)^2 \quad (9)$$

$$\times (\text{TF}(x, y, z_{\text{draft}}) + \delta T_{\text{sector}}) \quad (10)$$

$$170 \times |\langle \text{TF} \rangle_{\text{draft} \in \text{sector}} + \delta T_{\text{sector}}|, \quad (11)$$

where ρ_{sw} and c_{pw} are the density and specific heat capacity of seawater, ρ_i is the density of ice, L_f is the latent heat of
 fusion, and γ_0 is the ice-ocean heat flux. $\text{TF}(x, y, z_{\text{draft}})$ is the ocean thermal forcing anomaly at the depth of the ice shelf draft,
 calculated from ocean temperature and salinity outputs from CESM2-WACCM. Anomalies are calculated with respect to the
 initial thermal forcing such that basal mass balance is zero upon initialisation. Whilst this underestimates present-day melt rates
 175 in some key regions (Adusumilli et al., 2020), any errors are quickly outweighed by the strong climate forcing in SSP5-8.5 and
 we consider it to be adequate for the purposes of a sensitivity study. δT_{sector} is a sector-specific offset used by Jourdain et al.
 (2020) to fit basal melt to observed rates. However, our anomaly approach significantly reduces the impact of this term and we
 fix it constant at the ‘MeanAnt’ median-calibrated values derived by Jourdain et al. (2020).

Table 1. Perturbed parameters used in the ensemble.

Parameter	Long name	Min.	Nominal	Max.
α_C	C -regularisation	10^{-2}	10^0	10^2
α_ϕ	ϕ -regularisation	10^8	10^{10}	10^{12}
n	Flow law exponent	2	3	4
m	Sliding exponent	1	3	5
γ_0	Ice-ocean heat flux	9619	39131	159189

2.4 Experiment design

180 We assess sensitivity to parameters through a simple ‘axial’ ensemble, in which we perturb one parameter at a time to a given upper or lower bound whilst holding all other parameters at their chosen nominal value (Table 1). A method suggested by Wolovick et al. (2023) chooses a suitable range of regularisation values by considering the points at which L-curve curvature has dropped to a half of its maximum value (the maximum occurring at the L-curve corner). However, the authors note that the choice of a half is arbitrary and that the upper bound in particular is highly sensitive to this choice. In the interest of sampling

185 the widest possible uncertainty, we instead choose an upper and lower bound based on separate criteria: the lower bound is selected such that lower values of regularisation no longer improve the misfit (the solution is essentially unregularised); the upper bound corresponds to a solution which offers no (or marginal) improvement to the misfit over the initial guess. Note that, for C , this does not mean that the solution is equal to the driving stress, since smoother solutions can still improve the misfit. However, for ϕ , this results in a solution that is near-uniformly 1. This level of regularisation is well beyond the level

190 suggested by L-curve heuristics, but represents a case in which the model’s inability to capture processes normally absorbed by ϕ (macroscopic damage, crystal anisotropy, etc.) is laid bare.

For comparison with the regularisation parameters, we consider three other commonly perturbed ice sheet model parameters: the flow law exponent, n , the sliding exponent, m , and the ice-ocean heat flux, γ_0 . The nominal value of each parameter corresponds to either the arithmetic (m , n) or geometric (γ_0) mean of its upper and lower bounds. These are chosen to be

195 representative of typical values used across a range of studies (e.g. Nias et al., 2019; Barnes and Gudmundsson, 2022; Wolovick et al., 2023; Seroussi et al., 2024; Coulon et al., 2024, 2025; Rosier et al., 2025; Getraer and Morlighem, 2025). All in all, we perturb five parameters in BISICLES across a range of different model components. This gives us a total of 10 forward simulations from 6 initial states (some initial states are shared between ensemble members). This is a much simpler design than other space-filling options, which also allow for more sophisticated sensitivity analysis such as Sobol’ sensitivity (Hill

200 et al., 2021; Rosier et al., 2025). However, it is sufficient for a preliminary exploration of sensitivity before investing in a more comprehensive experiment design.

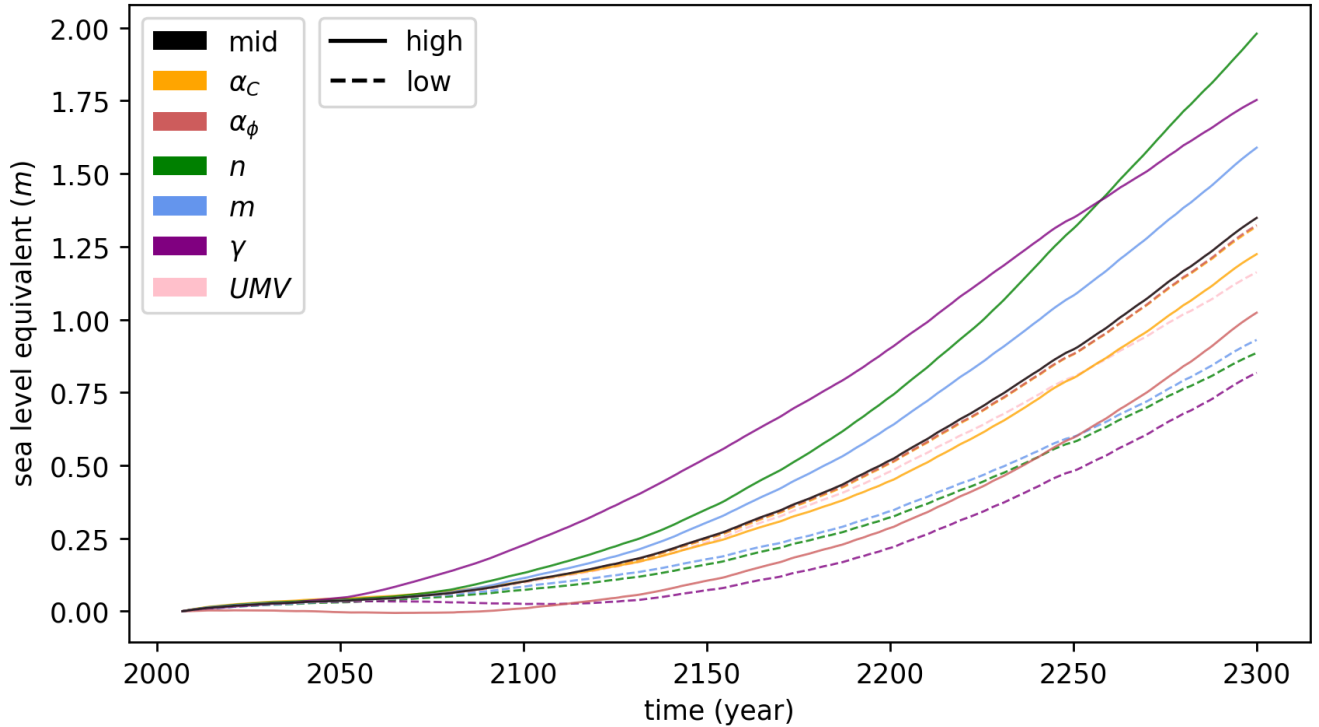


Figure 1. Timeseries of Antarctic sea level contribution for high and low values of each perturbed parameter.

3 Results

We project our initial states forward with the CESM2 climate forcing up to 2300 and calculate total Antarctic sea level contribution for each ensemble member (Figure 1). The ensemble range at 2100 is ??–?? m, which compares well with the IPCC
205 AR6 range of 0.03–0.28 m under SSP5–8.5 (cite IPCC). In the 22nd and 23rd centuries, all ensemble members accelerate their mass loss and the range widens to ??–?? m by 2300.

The spatial patterns of ice thickness change are broadly similar across the ensemble, with most mass loss occurring in regions of West Antarctica: the Amundsen Sea sector, the Siple coast, and the Weddell sea sector. However, the extent of mass loss in these regions depends on parameter choices, with high variance along Thwaites, Foundations, and Denman glaciers (Figure 2).
210 Some areas of low variance also show regions that retreat in all ensemble members, including the collapse of Pine Island Glacier and the Ross and Filchner-Ronne ice shelves.

For each parameter, we calculate the range in sea level contribution between ensemble members with minimal and maximal values of that parameter. This provides a rudimentary measure of sensitivity, which we can compare across parameters. Figure 3 highlights the timescale-dependance of this sensitivity, with different relative importance of parameters at 2100 and 2300.

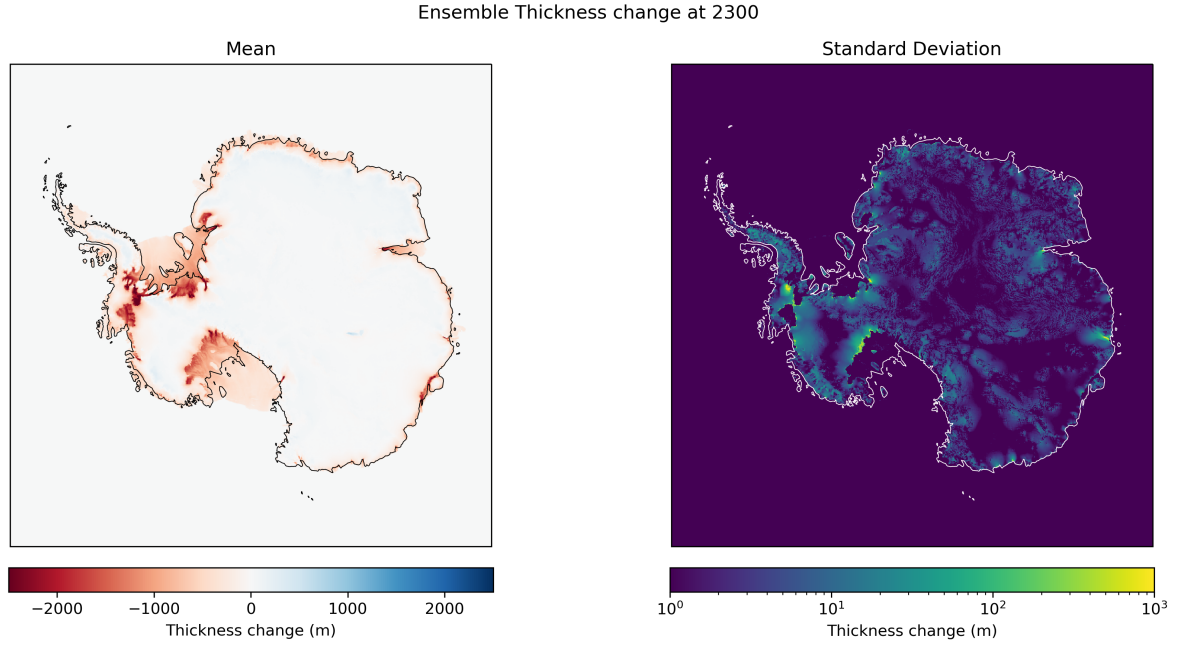


Figure 2. Spatial patterns of ice thickness change for high and low values of each perturbed parameter.

At 2100, the largest sensitivity is to ice shelf basal melt, γ_0 , but this reduces in relative terms by 2300 as other parameters become more important. Sensitivity to the flow law exponent, n , and the basal sliding exponent, m , both increase over time and are comparable to γ_0 by 2300. Sensitivities to glacial isostatic adjustment (UMV) and bed friction regularisation (a_C) are negligible at 2100 and remain small by 2300. Sensitivity to regularisation of the viscosity coefficient, a_ϕ , is the second largest of all parameters by 2100 but becomes less significant relative to other parameters by 2300.

4 Discussion

Sensitivity is sensitive to chosen range of values.

Both Martin et al. (2026) and Getraer and Morlighem (2025) show that Antarctic sea level projections can be very sensitive to the choice of n over multi-century timescales.

This is a much more rudimentary measure of sensitivity than, for example, Sobol' sensitivity (Hill et al., 2021; Rosier et al., 2025), and it doesn't provide any information about the shape of the relationship between parameters and outputs, as would be possible with a full perturbed parameter ensemble. However, it requires far fewer simulations and is sufficient for a preliminary exploration of sensitivity before investing in a more comprehensive experiment design.

The most direct comparison for our results is with (cite Rosier), who simulate the Amundsen Sea Embayment (ASE) up to 2100 using the $\tilde{U}$ a ice sheet model. Inversions in $\tilde{U}$ a are similar to those in BISICLES, using the adjoint method to solve for

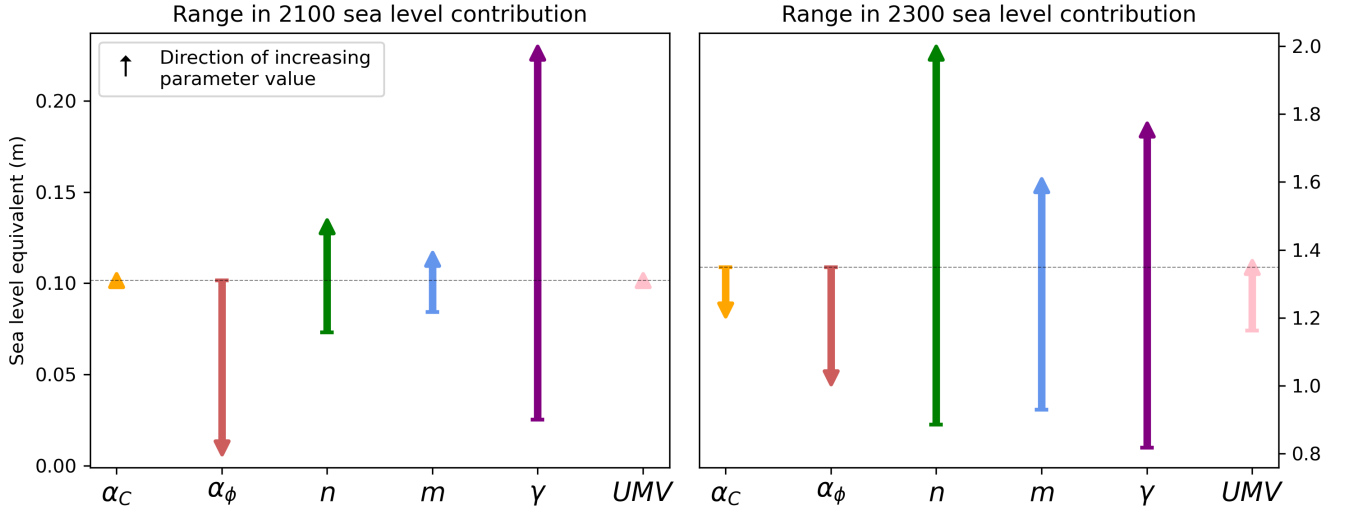


Figure 3. Sensitivity of Antarctic sea level contribution to each perturbed parameter at (a) 2100 and (b) 2300. Arrows denote the range in sea level contribution between ensemble members with minimal and maximal values of that parameter, pointing from minimal to maximal. Arrows pointing upwards/downwards therefore indicate a parameter that increases/decreases sea level contribution with higher values. The dashed horizontal line indicates the sea level contribution in a simulation with all parameters held at their default value.

230 basal friction and the rate factor, which is analogous to the viscosity coefficient in BISICLES insofar as it controls ice rheology. Similar to Rosier et al., we find that regularisation parameters in the initialisation can be a significant source of uncertainty in century-scale sea level projections. However, there are some notable differences between our results.

Both studies agree that sensitivity to ‘rheology-related’ regularisation (rate factor in $\dot{U}_a$, viscosity coefficient in BISICLES) is higher than sensitivity to bed friction regularisation. In BISICLES, the viscosity coefficient plays a critical role in dynamics
 235 near to the present-day grounding line, which are highly influential in short-term sea level projections. However, we find that this sensitivity decreases on longer timescales, possibly because the ice sheet retreats past present-day grounding lines into regions where the viscosity coefficient is mostly uniform.

Basal friction plays a role throughout the ice sheet and particularly in the ice streams, which may explain why sensitivity to its regularisation increases relative to rheology-related regularisation on longer timescales. However, sensitivity to bed friction
 240 regularisation in BISICLES is much lower than in $\dot{U}_a$ when compared against other commonly perturbed parameters. This may be due to differences in the domain and resolution of the models. The inclusion of slow-moving parts of East Antarctica in our simulations introduces very high friction values that may dominate the Tikhonov penalty function, focusing the smoothing on areas of Antarctica that are less dynamically important to century-scale sea level projections. BISICLES is also known to be sensitive to grid resolution up to at least 500 m (cite steph 2016), and our 1 km resolution may be too coarse to fully capture
 245 the influence of bed friction regularisation on ice dynamics.

The timescale dependence of our results highlight the importance of considering the projection period and forcing when assessing sources of uncertainty. Because of the very high CESM2-WACCM forcing post-2100, the major ice shelves collapse in all ensemble members before 2300, which reduces sensitivity to ice shelf basal melt relative to other parameters on this timescale. Similarly, the collapse of Pine Island Glacier in all ensemble members affects the sensitivity of our outputs to key parameters in that region, which could include bed friction regularisation. If we had used a less extreme forcing — either a lower-emission scenario or a different climate model — we may have found different patterns of sensitivity. Rosier et al find that parameters generally have higher sensitivity in a high-emissions scenario compared to a low-emissions scenario, but that this pattern is often reversed for initialisation parameters, supporting this idea.

5 Conclusions

We have explored the sensitivity of Antarctic sea level contribution to regularisation hyperparameters in the inverse problem and compared it with other commonly perturbed parameters. Using the BISICLES ice sheet model, the inverse problem solves for a spatially-varying friction coefficient (C) and viscosity coefficient (ϕ). We find that sensitivity to friction-regularisation is much lower than in an equivalent study for the Amundsen Sea Embayment and suggest that this is due to a difference in model domains and resolution, particularly the inclusion of large slow-moving regions of East Antarctica in our experiment. We find that sensitivity to viscosity-regularisation is high relative to other parameters on a century-timescale, but that this sensitivity declines over time as dynamical parameters dominate the multi-century response. This remains a valid approach to exploring initial state uncertainty in modelling experiments that initialise using inverse methods. However, we advise future experiments of this kind to consider the model, domain, resolution, and timescale of their experiment before choosing parameters to perturb, and warn that even if these parameters are included, it may not even then fully sample the uncertainty associated with the inverse problem.

Code availability. TEXT

Data availability. TEXT

Code and data availability. TEXT

Sample availability. TEXT

270 *Video supplement.* TEXT

Author contributions. TEXT

Competing interests. TEXT

Disclaimer. TEXT

Acknowledgements. TEXT

- Adusumilli, S., Fricker, H. A., Medley, B., Padman, L., and Siegfried, M. R.: Interannual Variations in Meltwater Input to the Southern Ocean from Antarctic Ice Shelves, *Nature Geoscience*, 13, 616–620, <https://doi.org/10.1038/s41561-020-0616-z>, 2020.
- Agosta, C., Amory, C., Kittel, C., Orsi, A., Favier, V., Gallée, H., Van Den Broeke, M. R., Lenaerts, J. T. M., Van Wessem, J. M., Van De Berg, W. J., and Fettweis, X.: Estimation of the Antarctic Surface Mass Balance Using the Regional Climate Model MAR (1979–2015) and
 280 Identification of Dominant Processes, *The Cryosphere*, 13, 281–296, <https://doi.org/10.5194/tc-13-281-2019>, 2019.
- Arthern, R. J., Hindmarsh, R. C. A., and Williams, C. R.: Flow Speed within the Antarctic Ice Sheet and Its Controls Inferred from Satellite Observations, *Journal of Geophysical Research: Earth Surface*, 120, 1171–1188, <https://doi.org/10.1002/2014JF003239>, 2015.
- Babaniyi, O., Nicholson, R., Villa, U., and Petra, N.: Inferring the Basal Sliding Coefficient Field for the Stokes Ice Sheet Model under Rheological Uncertainty, *The Cryosphere*, 15, 1731–1750, <https://doi.org/10.5194/tc-15-1731-2021>, 2021.
- 285 Barnes, J. M. and Gudmundsson, G. H.: The Predictive Power of Ice Sheet Models and the Regional Sensitivity of Ice Loss to Basal Sliding Parameterisations: A Case Study of Pine Island and Thwaites Glaciers, West Antarctica, *The Cryosphere*, 16, 4291–4304, <https://doi.org/10.5194/tc-16-4291-2022>, 2022.
- Barnes, J. M., Dias Dos Santos, T., Goldberg, D., Gudmundsson, G. H., Morlighem, M., and De Rydt, J.: The Transferability of Adjoint Inversion Products between Different Ice Flow Models, *The Cryosphere*, 15, 1975–2000, <https://doi.org/10.5194/tc-15-1975-2021>, 2021.
- 290 Berends, C. J., Van De Wal, R. S. W., Van Den Akker, T., and Lipscomb, W. H.: Compensating Errors in Inversions for Subglacial Bed Roughness: Same Steady State, Different Dynamic Response, *The Cryosphere*, 17, 1585–1600, <https://doi.org/10.5194/tc-17-1585-2023>, 2023.
- Bevan, S., Cornford, S., Gilbert, L., Otosaka, I., Martin, D., and Surawy-Stepney, T.: Amundsen Sea Embayment Ice-Sheet Mass-Loss Predictions to 2050 Calibrated Using Observations of Velocity and Elevation Change, *Journal of Glaciology*, 69, 1729–1739, <https://doi.org/10.1017/jog.2023.57>, 2023.
- 295 Brinkerhoff, D. J.: Variational Inference at Glacier Scale, *Journal of Computational Physics*, 459, 111095, <https://doi.org/10.1016/j.jcp.2022.111095>, 2022.
- Bulthuis, K., Arnst, M., Sun, S., and Pattyn, F.: Uncertainty Quantification of the Multi-Centennial Response of the Antarctic Ice Sheet to Climate Change, *The Cryosphere*, 13, 1349–1380, <https://doi.org/10.5194/tc-13-1349-2019>, 2019.
- 300 Cornford, S. L., Martin, D. F., Graves, D. T., Ranken, D. F., Le Brocq, A. M., Gladstone, R. M., Payne, A. J., Ng, E. G., and Lipscomb, W. H.: Adaptive Mesh, Finite Volume Modeling of Marine Ice Sheets, *Journal of Computational Physics*, 232, 529–549, <https://doi.org/10.1016/j.jcp.2012.08.037>, 2013.
- Cornford, S. L., Martin, D. F., Payne, A. J., Ng, E. G., Le Brocq, A. M., Gladstone, R. M., Edwards, T. L., Shannon, S. R., Agosta, C., van den Broeke, M. R., Hellmer, H. H., Krinner, G., Ligtenberg, S. R. M., Timmermann, R., and Vaughan, D. G.: Century-Scale Simulations of the
 305 Response of the West Antarctic Ice Sheet to a Warming Climate, *The Cryosphere*, 9, 1579–1600, <https://doi.org/10.5194/tc-9-1579-2015>, 2015.
- Cornford, S. L., Martin, D. F., Lee, V., Payne, A. J., and Ng, E. G.: Adaptive Mesh Refinement versus Subgrid Friction Interpolation in Simulations of Antarctic Ice Dynamics, *Annals of Glaciology*, 57, 1–9, <https://doi.org/10.1017/aog.2016.13>, 2016.
- Coulon, V., Klose, A. K., Kittel, C., Edwards, T., Turner, F., Winkelmann, R., and Pattyn, F.: Disentangling the Drivers of Future Antarctic
 310 Ice Loss with a Historically Calibrated Ice-Sheet Model, *The Cryosphere*, 18, 653–681, <https://doi.org/10.5194/tc-18-653-2024>, 2024.

- Coulon, V., Klose, A. K., Edwards, T., Turner, F., Pattyn, F., and Winkelmann, R.: From Short-Term Uncertainties to Long-Term Certainties in the Future Evolution of the Antarctic Ice Sheet, *Nature Communications*, 16, 10385, <https://doi.org/10.1038/s41467-025-66178-w>, 2025.
- Cuffey, K. M. and Paterson, W. S. B.: *The Physics of Glaciers*, Elsevier, San Diego, fourth edition edn., ISBN 978-0-12-369461-4 978-0-08-091912-6, 2010.
- Danabasoglu, G., Lamarque, J.-F., Bacmeister, J., Bailey, D. A., DuVivier, A. K., Edwards, J., Emmons, L. K., Fasullo, J., Garcia, R., Gettelman, A., Hannay, C., Holland, M. M., Large, W. G., Lauritzen, P. H., Lawrence, D. M., Lenaerts, J. T. M., Lindsay, K., Lipscomb, W. H., Mills, M. J., Neale, R., Oleson, K. W., Otto-Bliesner, B., Phillips, A. S., Sacks, W., Tilmes, S., Kampenhout, L., Vertenstein, M., Bertini, A., Dennis, J., Deser, C., Fischer, C., Fox-Kemper, B., Kay, J. E., Kinnison, D., Kushner, P. J., Larson, V. E., Long, M. C., Mickelson, S., Moore, J. K., Nienhouse, E., Polvani, L., Rasch, P. J., and Strand, W. G.: The Community Earth System Model Version 2 (CESM2), *Journal of Advances in Modeling Earth Systems*, 12, <https://doi.org/10.1029/2019MS001916>, 2020.
- DeConto, R. M., Pollard, D., Alley, R. B., Velicogna, I., Gasson, E., Gomez, N., Sadai, S., Condrón, A., Gilford, D. M., Ashe, E. L., Kopp, R. E., Li, D., and Dutton, A.: The Paris Climate Agreement and Future Sea-Level Rise from Antarctica, *Nature*, 593, 83–89, <https://doi.org/10.1038/s41586-021-03427-0>, 2021.
- Fox-Kemper, B., Hewitt, H. T., Xiao, C., Aðalgeirsdóttir, G., Drijfhout, S. S., Edwards, T. L., Golledge, N. R., Hemer, M., Kopp, R. E., Krinner, G., Mix, A., Notz, D., Nowicki, S., Nurhati, I. S., Ruiz, L., Sallée, J.-B., Slangen, A. B. A., and Yu, Y.: Ocean, Cryosphere and Sea Level Change, in: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, pp. 1211–1362, Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, <https://doi.org/10.1017/9781009157896.011>, 2021.
- Gagliardini, O., Zwinger, T., Gillet-Chaulet, F., Durand, G., Favier, L., De Fleurian, B., Greve, R., Malinen, M., Martín, C., Råback, P., Ruokolainen, J., Sacchetti, M., Schäfer, M., Seddik, H., and Thies, J.: Capabilities and Performance of Elmer/Ice, a New-Generation Ice Sheet Model, *Geoscientific Model Development*, 6, 1299–1318, <https://doi.org/10.5194/gmd-6-1299-2013>, 2013.
- Getrauer, B. and Morlighem, M.: Increasing the Glen–Nye Power-Law Exponent Accelerates Ice-Loss Projections for the Amundsen Sea Embayment, West Antarctica, *Geophysical Research Letters*, 52, e2024GL112516, <https://doi.org/10.1029/2024GL112516>, 2025.
- Glen, J. W.: The Creep of Polycrystalline Ice, *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 228, 519–538, <https://doi.org/10.1098/rspa.1955.0066>, 1955.
- Golledge, N. R., Kowalewski, D. E., Naish, T. R., Levy, R. H., Fogwill, C. J., and Gasson, E. G. W.: The Multi-Millennial Antarctic Commitment to Future Sea-Level Rise, *Nature*, 526, 421–425, <https://doi.org/10.1038/nature15706>, 2015.
- Gopalan, G., Hrafnkelsson, B., Aðalgeirsdóttir, G., and Pálsson, F.: Bayesian Inference of Ice Softness and Basal Sliding Parameters at Langjökull, *Frontiers in Earth Science*, 9, 610069, <https://doi.org/10.3389/feart.2021.610069>, 2021.
- Greve, R., Chambers, C., Obase, T., Saito, F., Chan, W.-L., and Abe-Ouchi, A.: Future Projections for the Antarctic Ice Sheet until the Year 2300 with a Climate-Index Method, *Journal of Glaciology*, pp. 1–11, <https://doi.org/10.1017/jog.2023.41>, 2023.
- Hansen, P. C.: Analysis of Discrete Ill-Posed Problems by Means of the L-Curve, *SIAM Review*, 34, 561–580, <https://doi.org/10.1137/1034115>, 1992.
- Hausfather, Z.: An Assessment of Current Policy Scenarios over the 21st Century and the Reduced Plausibility of High-Emissions Pathways, *Dialogues on Climate Change*, 2, 26–32, <https://doi.org/10.1177/29768659241304854>, 2025.
- Hill, E. A., Rosier, S. H. R., Gudmundsson, G. H., and Collins, M.: Quantifying the Potential Future Contribution to Global Mean Sea Level from the Filchner–Ronne Basin, Antarctica, *The Cryosphere*, 15, 4675–4702, <https://doi.org/10.5194/tc-15-4675-2021>, 2021.

- Hinkel, J., Church, J. A., Gregory, J. M., Lambert, E., Le Cozannet, G., Lowe, J., McInnes, K. L., Nicholls, R. J., Van Der Pol, T. D., and
 350 Van De Wal, R.: Meeting User Needs for Sea Level Rise Information: A Decision Analysis Perspective, *Earth's Future*, 7, 320–337,
<https://doi.org/10.1029/2018EF001071>, 2019.
- Hoffman, M. J., Perego, M., Price, S. F., Lipscomb, W. H., Zhang, T., Jacobsen, D., Tezaur, I., Salinger, A. G., Tuminaro, R., and Bertagna, L.:
 MPAS-Albany Land Ice (MALI): A Variable-Resolution Ice Sheet Model for Earth System Modeling Using Voronoi Grids, *Geoscientific
 Model Development*, 11, 3747–3780, <https://doi.org/10.5194/gmd-11-3747-2018>, 2018.
- 355 Isaac, T., Petra, N., Stadler, G., and Ghattas, O.: Scalable and Efficient Algorithms for the Propagation of Uncertainty from Data through
 Inference to Prediction for Large-Scale Problems, with Application to Flow of the Antarctic Ice Sheet, *Journal of Computational Physics*,
 296, 348–368, <https://doi.org/10.1016/j.jcp.2015.04.047>, 2015.
- Joughin, I., Smith, B. E., and Schoof, C. G.: Regularized Coulomb Friction Laws for Ice Sheet Sliding: Application to Pine Island Glacier,
 Antarctica, *Geophysical Research Letters*, 46, 4764–4771, <https://doi.org/10.1029/2019GL082526>, 2019.
- 360 Jourdain, N. C., Asay-Davis, X., Hattermann, T., Straneo, F., Seroussi, H., Little, C. M., and Nowicki, S.: A Protocol for Calculating Basal
 Melt Rates in the ISMIP6 Antarctic Ice Sheet Projections, *The Cryosphere*, 14, 3111–3134, <https://doi.org/10.5194/tc-14-3111-2020>,
 2020.
- Kopp, R. E., Gilmore, E. A., Little, C. M., Lorenzo-Trueba, J., Ramenzoni, V. C., and Sweet, W. V.: Usable Science for Managing the Risks
 of Sea-Level Rise, *Earth's Future*, 7, 1235–1269, <https://doi.org/10.1029/2018EF001145>, 2019.
- 365 Krishna, M., Morlighem, M., Mangini, D., and Choi, Y.: Quantifying the Uncertainty in Greenland's Contribution to Sea Level Rise Due to
 the Bed Topography, <https://doi.org/10.5194/egusphere-egu26-5996>, 2026.
- Le Clec'H, S., Quiquet, A., Charbit, S., Dumas, C., Kageyama, M., and Ritz, C.: A Rapidly Converging Initialisation Method to Simulate the
 Present-Day Greenland Ice Sheet Using the GRISLI Ice Sheet Model (Version 1.3), *Geoscientific Model Development*, 12, 2481–2499,
<https://doi.org/10.5194/gmd-12-2481-2019>, 2019.
- 370 MacAyeal, D. R.: The Basal Stress Distribution of Ice Stream E, Antarctica, Inferred by Control Methods, *Journal of Geophysical Research:*
Solid Earth, 97, 595–603, <https://doi.org/10.1029/91JB02454>, 1992.
- Martin, D. F., Kachuck, S. B., Trevers, M., Millstein, J. D., Cornford, S. L., and Minchew, B. M.: Rates of Sea-Level Rise Are Highly Sensitive
 to Ice Viscosity Parameters in Model Benchmarks, *AGU Advances*, 7, e2025AV001946, <https://doi.org/10.1029/2025AV001946>, 2026.
- Meinshausen, M., Nicholls, Z. R. J., Lewis, J., Gidden, M. J., Vogel, E., Freund, M., Beyerle, U., Gessner, C., Nauels, A., Bauer, N., Canadell,
 375 J. G., Daniel, J. S., John, A., Krummel, P. B., Luderer, G., Meinshausen, N., Montzka, S. A., Rayner, P. J., Reimann, S., Smith, S. J., van
 den Berg, M., Velders, G. J. M., Vollmer, M. K., and Wang, R. H. J.: The Shared Socio-Economic Pathway (SSP) Greenhouse Gas
 Concentrations and Their Extensions to 2500, *Geoscientific Model Development*, 13, 3571–3605, <https://doi.org/10.5194/gmd-13-3571-2020>, 2020.
- Morlighem, M.: MEaSURES BedMachine Antarctica, Version 3, <https://doi.org/10.5067/FPSU0V1MWUB6>, 2022.
- 380 Morlighem, M., Seroussi, H., Larour, E., and Rignot, E.: Inversion of Basal Friction in Antarctica Using Exact and Incomplete Adjoints of a
 Higher-order Model, *Journal of Geophysical Research: Earth Surface*, 118, 1746–1753, <https://doi.org/10.1002/jgrf.20125>, 2013.
- Nias, I. J., Cornford, S. L., and Payne, A. J.: Contrasting the Modelled Sensitivity of the Amundsen Sea Embayment Ice Streams, *Journal of
 Glaciology*, 62, 552–562, <https://doi.org/10.1017/jog.2016.40>, 2016.
- Nias, I. J., Cornford, S. L., Edwards, T. L., Gourmelen, N., and Payne, A. J.: Assessing Uncertainty in the Dynamical Ice Re-
 385 sponse to Ocean Warming in the Amundsen Sea Embayment, West Antarctica, *Geophysical Research Letters*, 46, 11 253–11 260,
<https://doi.org/10.1029/2019GL084941>, 2019.

- Nowicki, S., Goelzer, H., Seroussi, H., Payne, A. J., Lipscomb, W. H., Abe-Ouchi, A., Agosta, C., Alexander, P., Asay-Davis, X. S., Barthel, A., Bracegirdle, T. J., Cullather, R., Felikson, D., Fettweis, X., Gregory, J. M., Hattermann, T., Jourdain, N. C., Kuipers Munneke, P., Larour, E., Little, C. M., Morlighem, M., Nias, I., Shepherd, A., Simon, E., Slater, D., Smith, R. S., Straneo, F., Trusel, L. D., van den Broeke, M. R., and van de Wal, R.: Experimental Protocol for Sea Level Projections from ISMIP6 Stand-Alone Ice Sheet Models, *The Cryosphere*, 14, 2331–2368, <https://doi.org/10.5194/tc-14-2331-2020>, 2020.
- Nowicki, S. M. J., Payne, A., Larour, E., Seroussi, H., Goelzer, H., Lipscomb, W., Gregory, J., Abe-Ouchi, A., and Shepherd, A.: Ice Sheet Model Intercomparison Project (ISMIP6) Contribution to CMIP6, *Geoscientific Model Development*, 9, 4521–4545, <https://doi.org/10.5194/gmd-9-4521-2016>, 2016.
- O'Neill, J. F., Edwards, T. L., Martin, D. F., Shafer, C., Cornford, S. L., Seroussi, H. L., Nowicki, S., Adhikari, M., and Gregoire, L. J.: ISMIP6-based Antarctic Projections to 2100: Simulations with the BISICLES Ice Sheet Model, *The Cryosphere*, 19, 541–563, <https://doi.org/10.5194/tc-19-541-2025>, 2025.
- Petra, N., Martin, J., Stadler, G., and Ghattas, O.: A Computational Framework for Infinite-Dimensional Bayesian Inverse Problems, Part II: Stochastic Newton MCMC with Application to Ice Sheet Flow Inverse Problems, *SIAM Journal on Scientific Computing*, 36, A1525–A1555, <https://doi.org/10.1137/130934805>, 2014.
- Pollard, D. and DeConto, R. M.: A Simple Inverse Method for the Distribution of Basal Sliding Coefficients under Ice Sheets, *Applied to Antarctica*, *The Cryosphere*, 6, 953–971, <https://doi.org/10.5194/tc-6-953-2012>, 2012.
- Rignot, E., Mouginot, J., and Scheuchl, B.: MEaSUREs InSAR-Based Antarctica Ice Velocity Map, Version 2, <https://doi.org/10.5067/D7GK8F5J8M8R>, 2017.
- Rosier, S. H. R., Gudmundsson, G. H., Jenkins, A., and Naughten, K. A.: Calibrated Sea Level Contribution from the Amundsen Sea Sector, West Antarctica, under RCP8.5 and Paris 2C Scenarios, *The Cryosphere*, 19, 2527–2557, <https://doi.org/10.5194/tc-19-2527-2025>, 2025.
- Schoof, C. and Hindmarsh, R. C. A.: Thin-Film Flows with Wall Slip: An Asymptotic Analysis of Higher Order Glacier Flow Models, *The Quarterly Journal of Mechanics and Applied Mathematics*, 63, 73–114, <https://doi.org/10.1093/qjmam/hbp025>, 2010.
- Seroussi, H., Nowicki, S., Simon, E., Abe-Ouchi, A., Albrecht, T., Brondex, J., Cornford, S., Dumas, C., Gillet-Chaulet, F., Goelzer, H., Golledge, N. R., Gregory, J. M., Greve, R., Hoffman, M. J., Humbert, A., Huybrechts, P., Kleiner, T., Larour, E., Leguy, G., Lipscomb, W. H., Lowry, D., Mengel, M., Morlighem, M., Pattyn, F., Payne, A. J., Pollard, D., Price, S. F., Quiquet, A., Reerink, T. J., Reese, R., Rodehacke, C. B., Schlegel, N.-J., Shepherd, A., Sun, S., Sutter, J., Van Breedam, J., van de Wal, R. S. W., Winkelmann, R., and Zhang, T.: initMIP-Antarctica: An Ice Sheet Model Initialization Experiment of ISMIP6, *The Cryosphere*, 13, 1441–1471, <https://doi.org/10.5194/tc-13-1441-2019>, 2019.
- Seroussi, H., Verjans, V., Nowicki, S., Payne, A. J., Goelzer, H., Lipscomb, W. H., Abe-Ouchi, A., Agosta, C., Albrecht, T., Asay-Davis, X., Barthel, A., Calov, R., Cullather, R., Dumas, C., Galton-Fenzi, B. K., Gladstone, R., Golledge, N. R., Gregory, J. M., Greve, R., Hattermann, T., Hoffman, M. J., Humbert, A., Huybrechts, P., Jourdain, N. C., Kleiner, T., Larour, E., Leguy, G. R., Lowry, D. P., Little, C. M., Morlighem, M., Pattyn, F., Pelle, T., Price, S. F., Quiquet, A., Reese, R., Schlegel, N.-J., Shepherd, A., Simon, E., Smith, R. S., Straneo, F., Sun, S., Trusel, L. D., Van Breedam, J., Van Katwyk, P., Van De Wal, R. S. W., Winkelmann, R., Zhao, C., Zhang, T., and Zwinger, T.: Insights into the Vulnerability of Antarctic Glaciers from the ISMIP6 Ice Sheet Model Ensemble and Associated Uncertainty, *The Cryosphere*, 17, 5197–5217, <https://doi.org/10.5194/tc-17-5197-2023>, 2023.
- Seroussi, H., Pelle, T., Lipscomb, W. H., Abe-Ouchi, A., Albrecht, T., Alvarez-Solas, J., Asay-Davis, X., Barre, J.-B., Berends, C. J., Bernales, J., Blasco, J., Caillet, J., Chandler, D. M., Coulon, V., Cullather, R., Dumas, C., Galton-Fenzi, B. K., Garbe, J., Gillet-Chaulet, F., Gladstone, R., Goelzer, H., Golledge, N., Greve, R., Gudmundsson, G. H., Han, H. K., Hillebrand, T. R., Hoffman, M. J., Huybrechts, P.,

- 425 Jourdain, N. C., Klose, A. K., Langebroek, P. M., Leguy, G. R., Lowry, D. P., Mathiot, P., Montoya, M., Morlighem, M., Nowicki, S.,
Pattyn, F., Payne, A. J., Quiquet, A., Reese, R., Robinson, A., Saraste, L., Simon, E. G., Sun, S., Twarog, J. P., Trusel, L. D., Urruty, B.,
Van Breedam, J., Van De Wal, R. S. W., Wang, Y., Zhao, C., and Zwinger, T.: Evolution of the Antarctic Ice Sheet Over the Next Three
Centuries From an ISMIP6 Model Ensemble, *Earth's Future*, 12, e2024EF004561, <https://doi.org/10.1029/2024EF004561>, 2024.
- Shewchuk, J. R.: An Introduction to the Conjugate Gradient Method without the Agonizing Pain, Tech. rep., Carnegie Mellon University,
430 USA, 1994.
- Siahaan, A., Smith, R. S., Holland, P. R., Jenkins, A., Gregory, J. M., Lee, V., Mathiot, P., Payne, A. J., Ridley, J. K., and Jones, C. G.:
The Antarctic Contribution to 21st-Century Sea-Level Rise Predicted by the UK Earth System Model with an Interactive Ice Sheet, *The
Cryosphere*, 16, 4053–4086, <https://doi.org/10.5194/tc-16-4053-2022>, 2022.
- Sun, S., Pattyn, F., Simon, E. G., Albrecht, T., Cornford, S., Calov, R., Dumas, C., Gillet-Chaulet, F., Goelzer, H., Golledge, N. R.,
435 Greve, R., Hoffman, M. J., Humbert, A., Kazmierczak, E., Kleiner, T., Leguy, G. R., Lipscomb, W. H., Martin, D., Morlighem, M.,
Nowicki, S., Pollard, D., Price, S., Quiquet, A., Seroussi, H., Schlemm, T., Sutter, J., van de Wal, R. S. W., Winkelmann, R., and
Zhang, T.: Antarctic Ice Sheet Response to Sudden and Sustained Ice-Shelf Collapse (ABUMIP), *Journal of Glaciology*, 66, 891–904,
<https://doi.org/10.1017/jog.2020.67>, 2020.
- Trevers, M., Cornford, S. L., Payne, A. J., Gasson, E., and Bevan, S. L.: Retreat of Thwaites Glacier Triggered by Its Neighbours,
440 <https://doi.org/10.22541/essoar.170957499.97583808/v1>, 2024.
- Wernecke, A., Edwards, T. L., Holden, P. B., Edwards, N. R., and Cornford, S. L.: Quantifying the Impact of Bedrock To-
pography Uncertainty in Pine Island Glacier Projections for This Century, *Geophysical Research Letters*, 49, e2021GL096589,
<https://doi.org/10.1029/2021GL096589>, 2022.
- Wolovick, M., Humbert, A., Kleiner, T., and Rückamp, M.: Regularization and L-curves in Ice Sheet Inverse Models: A Case Study in the
445 Filchner–Ronne Catchment, *The Cryosphere*, 17, 5027–5060, <https://doi.org/10.5194/tc-17-5027-2023>, 2023.